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Oefeningenreeks 1C nr. 46.5

Trek de vierdemachtswortels in $\mathbb{C}$:

$z^4 = -4$

We weten dat:

$$-4 = 4(-1) = 4(\cos(\pi) + i\sin(\pi)) = 4\operatorname{cis}(\pi)$$

$$\Rightarrow z_k = \sqrt{2}\operatorname{cis}\left(\frac{\pi + 2k\pi}{4}\right), k = 0, 1, 2, 3$$

$$\Rightarrow z_k = \sqrt{2}\operatorname{cis}\left(\frac{\pi}{4} + \frac{k\pi}{2}\right), k = 0, 1, 2, 3$$

$$\Rightarrow z_0 = \sqrt{2}\operatorname{cis}\left(\frac{\pi}{4}\right) = 1 + i$$

$$\Rightarrow z_1 = \sqrt{2}\operatorname{cis}\left(\frac{3\pi}{4}\right) = -1 + i$$

$$\Rightarrow z_2 = \sqrt{2}\operatorname{cis}\left(\frac{5\pi}{4}\right) = -1 - i$$

$$\Rightarrow z_3 = \sqrt{2}\operatorname{cis}\left(\frac{7\pi}{4}\right) = 1 - i$$

$$\Rightarrow \boxed{z_0 = 1 + i, z_1 = -1 + i, z_2 = -1 - i, z_3 = 1 - i}$$

References